Space Development Agency
Battle Management Multi-Mission Module Architecture for the Proliferated Warfighter Space Architecture
Request for Information

SDA-SN-25-0015

The Space Development Agency (SDA) seeks industry feedback on in-space Battle Management, Command, Control and Communications (BMC3) Multi-Mission Module (M3) interface documentation to inform SDA's future BMC3 M3 solicitations.

SDA is releasing this Request for Information (RFI) to assist in the development and refinement of "interface documentation" that will be used to govern the interface between a future M3 and the SDA spacecraft it resides on.

With this RFI, SDA is soliciting recommendations and modifications to the attached documentation. With compliance to this documentation, the M3s and their spacecraft interface are intended to allow SDA to apply a Modular Open System Approach (MOSA) to the development of future space edge processing hardware. This MOSA approach with allow component vendors to independently develop capabilities that are compliant to this interface documentation. These components will then be able to be integrated by space vehicle contractors whose interface is also compliant. All qualified components will then be available for space vehicle contractors to acquire and host on their vehicles when responding to contract requirements.

Background

SDA was created in March 2019 to provide fast, responsive, resilient solutions to national security space requirements. Its focus is on the Proliferated Warfighter Space Architecture (PWSA), an "information backbone" of hundreds of satellites in several layers networked together. These layers provide access to information supporting missile warning, air, ground, and surface object tracking, and other missile defense capabilities. The goal is to make data available to the warfighter "at or ahead of the speed of the threat."

Recognized as the Department of Defense's constructive disruptor for space acquisition, SDA will quickly deliver needed space-based capabilities to the joint warfighter to support terrestrial missions through development, fielding, and operation of the PWSA. SDA capitalizes on a unique business model that values speed and lowers costs by harnessing commercial development to achieve a proliferated architecture and enhance resiliency. Having interface control documentation that industry has contributed to is critical to making this transition.

SDA is responsible for orchestrating the Department's future threat-driven space architecture and accelerating the development and fielding of new military space capabilities necessary to ensure U.S. technological and military advantage in space for national defense. To achieve this mission, SDA will unify and integrate next-generation space capabilities to deliver the PWSA, a resilient military sensing and data transport capability via a proliferated space architecture primarily in Low Earth Orbit (LEO). Some of the reference missions for the M3 within the PWSA are hosting applications that support; network management, local network autonomy, advanced Red-Black routing, advanced local-distributed timing solutions, advanced sensor layer edge processing, 2D-to-3D tracking of missile threats, etc...

The SDA mission, aligned with the National Defense Strategy (NDS), defines and delivers the Department's future resilient, threat-driven, and affordable military space architecture for the joint warfighter. With the NDS and the capability gaps identified in the DoD Space Vision guiding our thinking, SDA will continuously refine and prioritize activities to ensure U.S. leadership in space. SDA's mission begins and ends with the warfighter. SDA recognizes that good enough capabilities in the hands of the warfighter sooner may be better than delivering the perfect solution too late. SDA remains mission-focused and continues to make advances toward delivering the next-generation PWSA.

Description

The SDA Battle Management layer will provide the hardware and software framework to host mission-specific processing, algorithms, and applications across the architecture layers and systems. Supported use cases will include space-based command and control, tasking, and mission processing. Processed data will be routed across a mesh network through cross-links and down-links to enable timely dissemination to both the warfighter and other systems in the architecture. Battle Management will continuously address evolving threats and mission needs through on-orbit updates to hardware/firmware with each tranche delivery and on-orbit updates to flight software and mission applications throughout the lifecycle of each tranche.

To meet these needs, future space-based edge processing hardware and spacecraft accommodations will need to comply with the provided interface documentation that we are seeking comments on. We have drafted this interface documentation to directly respond to DoD guidance to create development objectives that support MOSA for major subsystem design integration. In the spirit of true collaboration, **these interface definitions will be negotiable** through this RFI interaction and following interactions. The future of space processing will be evolving faster than anyone can predict. Having the ability to ensure design interoperability will allow the PWSA to be able to be as responsive as possible.

The two M3 interface documents that are the center of this exchange are included in Appendix A & B. A third document to address data formatting between the spacecraft and the M3 will be addressed in follow-on interactions.

SDA's Request for Information (RFI) Response Timeline

Comments and inputs for this RFI addressing interface documentation should be received by close of business on 11 Apr 2025.

Appendix A – Multi-Mission Module Interface Control Document
Appendix B – M3 Mechanical Interface Control Drawing March 2025